

One boundary extension with all five analytic properties

MathlibAnnex.Plucker.exists_extension_with_derivativeAverage_mem

theorem

Keeps boundary agreement, contraction, derivative control, integrability and average membership on one witness.

Statement

Let $n, N \in \mathbb{N}_0$ and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p : E \rightarrow \mathbb{R}$ be a continuous real seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in E).$$

The lower bound makes p a norm. Its closed unit ball and real reference volume are

$$K_p = \{x : p(x) \leq 1\}, \quad v_p = \lambda_n(K_p) \in (0, \infty).$$

Here and below a finite Lebesgue measure is read as a real number when used as a scalar. The boundary is the unit sphere $\Sigma_p = \{x : p(x) = 1\}$.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B)$ for that vector scaled by v_p . Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_p(B) = v_p \Pi_n(B) \quad (I = (i_1 < \dots < i_n)).$$

A p -contraction is a linear map $B : E \rightarrow F$ satisfying $\|Bz\|_\infty \leq p(z)$ for every $z \in E$. Write $\mathcal{C}_p^N$ for these maps. Write $\mathcal{G}_p^N$ for the signed generator set and $\mathcal{P}_p^N$ for its real convex hull, so

$$\mathcal{G}_p^N = \{\mathcal{V}_p(B), -\mathcal{V}_p(B) : B \in \mathcal{C}_p^N\}, \quad \mathcal{P}_p^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_p^N).$$

For any function $k : E \rightarrow F$, write $Dk(x)$ for its Fréchet derivative where it exists and for the zero linear map otherwise. Write $G_{p,k}$ for its derivative-generator function and $\mathcal{A}_p(k)$ for the average of that function over K_p . The integrand and normalization are

$$G_{p,k}(x) = v_p \Pi_n(Dk(x)), \quad \mathcal{A}_p(k) = v_p^{-1} \int_{K_p} G_{p,k}(x) d\lambda_n(x).$$

The integrals use the fixed reference measure, even when distances are measured with p .

Let $h : \Sigma_p \rightarrow F$ satisfy $\|h(u) - h(v)\|_\infty \leq p(u - v)$ for all $u, v \in \Sigma_p$. There is one function $f : E \rightarrow F$ with all five properties:

$$\begin{aligned}
&\|f(x) - f(y)\|_\infty \leq p(x - y) && (x, y \in E), \\
&f(u) = h(u) && (u \in \Sigma_p), \\
&\|Df(x)z\|_\infty \leq p(z) && (z \in E), \quad \lambda_n|_{K_p}\text{-a.e. } x, \\
&G_{p,f} \in L^1(K_p, \lambda_n), \\
&\mathcal{A}_p(f) \in \mathcal{P}_p^N.
\end{aligned}$$

Assumptions

The source and target are finite real coordinate spaces with sup norms. The seminorm p is continuous and has the stated positive lower and upper comparison constants. The measure is the original coordinate Lebesgue measure. The boundary map is 1-Lipschitz for the p -distance and the coordinate sup norm on F . No differentiability of h , nonempty sphere, or positive dimension is required. The dimensions n, N may be zero. When $n = 0$, the unique empty minor has determinant 1; when $N < n$, the minor-index set is empty and its function space has one element. Neither case is excluded.

Conclusion

The single extension f satisfies the displayed five properties simultaneously. Its average is taken over K_p with reference Lebesgue measure. This asserts existence of a common witness, not uniqueness.

The radial extension of a sphere isometry is a separate construction. This theorem does not identify its witness with a radial map.

Proof route

Extend coordinatewise in the induced norm, return to reference coordinates, then derive the remaining properties for that same function.

▼ Proof steps

1. **Construct an extension in the correct metric.** On the same vectors of E , use the distance $d_p(x, y) = p(x - y)$. If Σ_p is empty, take $f = 0$. Otherwise write h_i for coordinate i of h and, for $1 \leq i \leq N$, use

$$f_i(x) = \inf_{u \in \Sigma_p} (h_i(u) + p(x - u)).$$

Fix $u_0 \in \Sigma_p$. The boundary Lipschitz inequality and the triangle inequality give, for every $u \in \Sigma_p$,

$$h_i(u_0) - p(x - u_0) \leq h_i(u) + p(x - u).$$

The infimum is finite: this is a lower bound, and $u = u_0$ gives a finite upper bound. For $v \in \Sigma_p$ and $u \in \Sigma_p$,

$$h_i(v) \leq h_i(u) + p(v - u), \quad f_i(v) \leq h_i(v) + p(v - v) = h_i(v).$$

Taking the infimum in the first inequality yields $f_i(v) = h_i(v)$. For any $x, y \in E$, the triangle inequality in the defining formula gives

$$f_i(x) \leq f_i(y) + p(x - y), \quad |f_i(x) - f_i(y)| \leq p(x - y).$$

The second bound follows by also interchanging x and y . Hence the coordinate map $f = (f_i)_{i=1}^N$ satisfies

$$\|f(x) - f(y)\|_\infty \leq p(x - y), \quad f|_{\Sigma_p} = h.$$

For $N = 0$ the target is the zero space and these assertions still hold, without taking a maximum of an empty family. The scalar McShane construction and its finite-coordinate version prove precisely these extension properties. The source theorem chooses a witness of that result; all following steps use that one f , read in the original reference coordinates.

2. Pass this same extension to its derivative. The first output gives

$$\|f(x) - f(y)\|_\infty \leq p(x - y) \leq b_p \|x - y\|_\infty.$$

The reference-norm Lipschitz conversion supplies a global b_p -Lipschitz bound. Apply the almost-everywhere directional bound with the same f , seminorm p , upper constant b_p , and restricting set K_p . Its output is the third property, $\|Df(x)z\|_\infty \leq p(z)$ for all z , almost everywhere on K_p .

3. Prove integrability for this witness. Let $C \geq 0$ bound the compact generator set $\mathcal{G}_p^N$. The derivative bound just obtained gives

$$G_{p,f}(x) \in \mathcal{G}_p^N, \quad \|G_{p,f}(x)\| \leq C \quad \text{a.e. on } K_p,$$

$$\int_{K_p} \|G_{p,f}(x)\| d\lambda_n(x) \leq C v_p < \infty.$$

The total derivative is measurable; continuous volume-scaled minors make this finite-coordinate vector strongly measurable. Compactness of the signed generators supplies C , and the ball has finite measure. These are the actual inputs of integrability under the seminorm increment bound, which yields the fourth property for the same f .

4. Average without changing the witness. We now have

$$G_{p,f} \in L^1(K_p), \quad Df(x) \in \mathcal{C}_p^N \quad \text{a.e. on } K_p.$$

Apply derivative-average membership to exactly these two outputs. It gives $\mathcal{A}_p(f) \in \mathcal{P}_p^N$. Together with the two extension properties in Step 1, this is one existential witness with all five conclusions.

Main citations

- Exact declaration and its source — `MathlibAnnex.Plucker.exists_extension_with_derivativeAverage_mem`
- The finite-coordinate McShane extension theorem — `LipschitzOnWith.extend_pi`
- The reference-norm Lipschitz conversion — `MathlibAnnex.EquivalentSeminorm.lipschitzWith_upper_of_model_bound`
- the almost-everywhere directional bound — `MathlibAnnex.FDeriv.ae_norm_apply_le_seminorm_of_lipschitz`
- integrability under the seminorm increment bound — `MathlibAnnex.Plucker.integrableOn_derivativeGenerator_of_seminormLipschitz`
- derivative-average membership — `MathlibAnnex.Plucker.derivativeAverage_mem_body`

- The scalar McShane construction — `LipschitzOnWith.extend_real`
- Compactness of the signed generators — `MathlibAnnex.PluckerBody.isCompact_generators`
- Signed generator set — `MathlibAnnex.PluckerBody.generators`
- Convex-body definition — `MathlibAnnex.PluckerBody.body`
- Derivative-generator function — `MathlibAnnex.Plucker.derivativeGenerator`
- Set-average definition — `MathlibAnnex.Plucker.derivativeAverage`

Lean source signature (exact)

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/-- Every seminorm-Lipschitz boundary map has an extension with its prescribed
trace, almost everywhere contractive derivative, integrable derivative
generator, and derivative average in the Plücker body. -/
theorem exists_extension_with_derivativeAverage_mem {n N : ℕ}
  (M : EquivalentSeminorm (Fin n → ℝ)) (g : M.unitSphere → (Fin N → ℝ))
  (hg : ∀ u v, ‖g u - g v‖ ≤ M.p ((u : Fin n → ℝ) - (v : Fin n → ℝ))) :
  ∃ f : (Fin n → ℝ) → (Fin N → ℝ),
    (∀ x y, ‖f x - f y‖ ≤ M.p (x - y)) ∧
    (∀ u : M.unitSphere, f u = g u) ∧
    (∀m x ∂volume.restrict M.closedUnitBall, M.IsContraction (fderiv ℝ f x)) ∧
    IntegrableOn (derivativeGenerator M f) M.closedUnitBall volume ∧
    derivativeAverage M f ∈ PluckerBody.body M N

```

Before the final colon, M , g and hg are inputs. After it, $\exists f$ says that there exists **one function** $f : E \rightarrow F$ satisfying all five clauses joined by $\wedge$ (“and”).

In the source	Mathematical meaning
<code>{n N : ℕ}</code>	Arbitrary nonnegative integers n, N . Braces mean that Lean may infer these parameters; they are not extra assumptions.
<code>Fin n → ℝ; Fin N → ℝ</code>	The spaces $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ with their sup norms. A vector is a list of real coordinates, indexed from 0 in Lean and from 1 in the formulas here.
<code>M : EquivalentSeminorm (Fin n → ℝ)</code>	The continuous seminorm $p = M.p$, with $a_p = M.lower > 0$, $b_p = M.upper > 0$ and $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every $x \in E$.
<code>M.unitSphere</code>	The unit sphere $\Sigma_p = \{u \in E : p(u) = 1\}$.
<code>g : M.unitSphere → (Fin N → ℝ)</code>	The given boundary map $h : \Sigma_p \rightarrow F$. The source calls it g , while the mathematical text calls it h .
<code>(u : Fin n → ℝ); f u</code>	Regard the sphere point u as the same coordinate vector in E ; hence $f u$ means $f(u)$.
<code>hg : ∀ u v, ‖g u - g v‖ ≤ M.p (...)</code>	The boundary hypothesis $\ h(u) - h(v)\ _\infty \leq p(u - v)$ for all $u, v \in \Sigma_p$.

In the source	Mathematical meaning
<code>M.closedUnitBall; M.closedUnitBallVolume</code>	$K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda_n(K_p) > 0$, with the finite measure read as a real number.
<code>fderiv ℝ f x</code>	The real Fréchet derivative $Df(x) : E \rightarrow F$, with the zero linear map used when f is not differentiable at x .
<code>derivativeGenerator M f x</code>	The vector $G_{p,f}(x) = v_p \Pi_n(Df(x))$; its $I = (i_1 < \dots < i_n)$ coordinate is $v_p \det ((Df(x)e_j)_{a,j=1}^n)_{a,i=1}^n$.
<code>generators M N</code>	The set $\mathcal{G}_p^N = \{\pm v_p \Pi_n(B) : B : E \rightarrow F \text{ linear}, \forall z \in E, \ Bz\ _\infty \leq p(z)\}$. Its elements are minor-coordinate vectors, not the maps B .
<code>PluckerBody.body M N</code>	$\mathcal{P}_p^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_p^N)$: all finite convex combinations of the signed vectors in the generator set just described.
<code>derivativeAverage M f</code>	The vector $\mathcal{A}_p(f) = v_p^{-1} \int_{K_p} G_{p,f}(x) d\lambda_n(x)$, not the average of the linear maps $Df(x)$.

The five output clauses, in their source order, mean:

In the source	Mathematical meaning
$\forall x \ y, \ f \ x - f \ y\ \leq M.p \ (x - y)$	$\ f(x) - f(y)\ _\infty \leq p(x - y)$ for every $x, y \in E$.
$\forall u : M.\text{unitSphere}, f \ u = g \ u$	$f(u) = h(u)$ for every $u \in \Sigma_p$.
$\forall^m x \ \partial\text{volume.restrict } M.\text{closedUnitBall}, M.\text{IsContraction } (f\text{deriv } \mathbb{R} \ f \ x)$	For Lebesgue-almost every $x \in K_p$, $\ Df(x)z\ _\infty \leq p(z)$ for every direction $z \in E$. The exceptional null set does not depend on z .

In the source	Mathematical meaning
<code>IntegrableOn (derivativeGenerator M f) M.closedUnitBall volume</code>	The same $G_{p,f}$ is measurable up to a null set on K_p and $\int_{K_p} \ G_{p,f}(x)\ _\infty d\lambda_n(x) < \infty$.
<code>derivativeAverage M f ∈ PluckerBody.body M N</code>	The same extension satisfies $\mathcal{A}_p(f) \in \mathcal{P}_p^N$.

Here $\forall^m$ means “for almost every”, `volume.restrict M.closedUnitBall` is λ_n restricted to K_p , and $\in$ is set membership. These five lines are conclusions about the chosen f , not hypotheses imposed on the original boundary map h .

Exact definitions used in this reading: Signed generator set; Convex-body definition; Derivative-generator function; Set-average definition.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 14–19:

```
noncomputable section

open Set MeasureTheory
open MathlibAnnex.EquivalentSeminorm

namespace MathlibAnnex.Plucker
```

▼ Exact content identity

Declaration: MathlibAnnex.Plucker.exists_extension_with_derivativeAverage_mem

Accepted content SHA-256: 9880f3b568c873c74b81a9bb3688a98b29fb5dc53016872a2a1b71df9af71736

Accepted source guide SHA-256: 9083fd3b3ec1ac4401c7b1797cae71b933f8e92cac393b94f84e98594a6d25d9

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73